Prognostic Value of Right Ventricular Function for Heart Failure in Hypertrophic Cardiomyopathy with Latent Obstruction

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Background

In patients with hypertrophic cardiomyopathy (HCM), left ventricular outflow tract (LVOT) obstruction caused by septal contact of the mitral valve is a key driver of heart failure (HF), leading to limiting symptoms, impaired quality of life, and adverse prognosis. Because LVOT obstruction is strongly influenced by LV filling conditions, a subgroup of patients demonstrates latent obstruction that becomes apparent only during provocation or exercise. In these patients, right ventricular (RV) could play a critical role in biventricular coupling and maintaining LV filling. We therefore hypothesized that comprehensive RV functional analysis using CMR feature tracking could provide prognostic value for predicting HF events in HCM patients with latent LVOT obstruction.

Methods

In a large multicenter cohort of 1,333 patients with obstructive HCM who underwent CMR between 2003 and 2023, we identified 284 patients with NYHA class I/II symptoms, preserved LVEF (≥50%), and no prior septal reduction therapy (SRT), who had an LVOT gradient <30 mmHg at rest but ≥30 mmHg during provocation. Biventricular volumes and function were quantified, and LGE was manually assessed. Feature tracking analysis was performed on four-chamber cine images using QStrain (Medis 4.0.70.4, Leiden, The Netherlands) to derive RV global longitudinal strain (GLS) and free wall strain **(Figure 1)**.The primary endpoint was a composite of HF progression (NYHA class III/IV), new LVEF <50%, or SRT. Univariable Cox regression analyses

were performed including demographical, echocardiographic and CMR variables. Variables with $p<0.05$ were entered into multivariable Cox regression models. Kaplan–Meier analysis was used to visualize the prognostic relevance of RV strain, with the optimal cutoff determined by Youden's index.

Results

Over a mean follow-up of 5.0 ± 3.3 years, 69 patients (24%) experienced HF events. Compared with event-free patients, those with HF events were more often female, more frequently in NYHA class II at baseline, had higher peak LVOT gradients, and more often presented with mitral regurgitation. They were also more frequently treated with beta-blockers and demonstrated lower RV GLS and RV free wall strain, whereas no other differences in biventricular volumes or function were observed (**Table 1**). In regression analyses, RV GLS emerged as an independent predictor of HF events, alongside NYHA class II, greater severity of mitral regurgitation, and higher peak LVOT gradient (**Figure 2A**). At an optimal cutoff of –23%, patients with lower RV GLS had significantly higher probability of HF Events (**Figure 2B**).

Conclusion

In HCM patients with provocable LVOT obstruction, RV GLS as assessed by CMR feature tracking emerged as an independent predictor of adverse HF events, highlighting the importance of comprehensive evaluation of biventricular coupling in this population.

Table 1: Patient characteristics and CMR Parameters

Patient characteristics and medical history	No HF Event (n=215)	HF Event (n=69)	p-value

Sex (male)	**78.1% (168)**	**60.9% (42)**	**0.004**
Age at initial visit (years)	48.57 ± 16.62	49.91 ± 14.40	0.519
Body mass index (kg/m²)	29.29 ± 5.54	30.33 ± 5.41	0.183
NYHA Class II	**39.5% (85)**	**79.7% (55)**	**<0.001**
Atrial fibrillation	13.0% (28)	13.0% (9)	0.997
Arterial hypertension	38.6% (83)	33.3% (23)	0.431
Coronary artery disease	6.0% (13)	4.3% (3)	0.594
Family history of HCM	20.9% (45)	17.4% (12)	0.523
Positive record of genetic mutation	5.6% (12)	4.3% (3)	0.69
Apical aneurysm	1.4% (3)	0.0% (0)	0.324
Mitral regurgitation			
Grade 1	29.8% (64)	31.9% (22)	0.739
Grade 2	**1.9% (4)**	**11.6% (8)**	**0.001**
Grade 3	0 (0.0%)	0 (0.0%)	n/a
Grade 4	0 (0.0%)	0 (0.0%)	n/a
Peak LVOT Gradient			
30-49 mmHg	**30.7% (66)**	**15.9% (11)**	**0.016**
50-85 mmHg	43.3% (93)	40.6% (28)	0.696
>85 mmHg	**26.0% (56)**	**43.5% (30)**	**0.006**
Betablocker	**43.3% (93)**	**63.8% (44)**	**0.003**
Calcium Channel Blockers	19.5% (42)	18.8% (13)	0.899
ACE-Inhibitor/ ARB	22.3% (48)	18.8% (13)	0.54
Disopyramide	0.9% (2)	1.4% (1)	0.714
CMR			
Maximum WT (mm)	17.70 ± 4.15	17.83 ± 3.92	0.815
LV mass index (g/m²)	73.69 ± 23.59	73.83 ± 25.42	0.968
LA diameter (mm)	38.67 ± 8.52	39.61 ± 8.33	0.417
LV end diastolic volume index (ml/m²)	76.75 ± 16.41	75.36 ± 18.10	0.573
LV stroke volume index (ml/m²)	51.15 ± 11.04	51.29 ± 12.03	0.933
LV ejection fraction (%)	66.84 ± 6.67	68.12 ± 7.62	0.212
RV end diastolic volume index (ml/m²)	69.50 ± 17.52	66.60 ± 17.90	0.244
RV stroke volume index (ml/m²)	40.40 ± 10.85	39.08 ± 11.39	0.407

RV ejection fraction (%)	58.57 ± 8.38	58.72 ± 7.59	0.897
RV GLS (%)	**-24.57 ± 5.56**	**-26.79 ± 4.70**	**0.001**
RV Free Wall Strain (%)	**-31.82 ± 7.22**	**-34.34 ± 6.48**	**0.007**
LGE presence	46.0% (99)	43.5% (30)	0.709
Relative LGE mass (%)	**2.19 ± 4.04**	**1.28 ± 2.26**	**0.022**

WT – Wall thickness, LVOT – Left ventricular outflow tract, ARB – Angiotensin II receptor blocker, LV/RV – Left/Right ventricle, LA – Left atrium, GLS – Global longitudinal strain, LGE – Late gadolinium enhancement

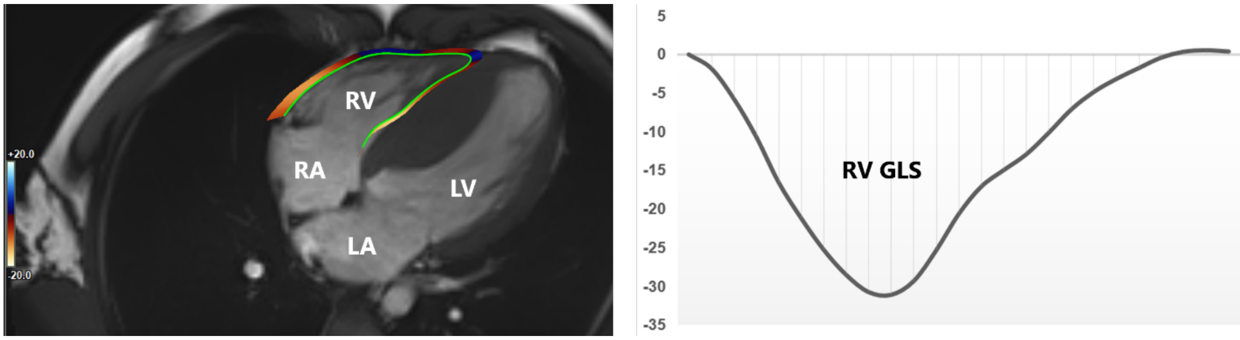

Figure 1. Example of right ventricular feature tracking analysis in a patient with hypertrophic cardiomyopathy.
Left: RV contouring for feature tracking analysis.
Right: Corresponding RV strain curve.
RA/LA = right/left atrium; LV = left ventricle.

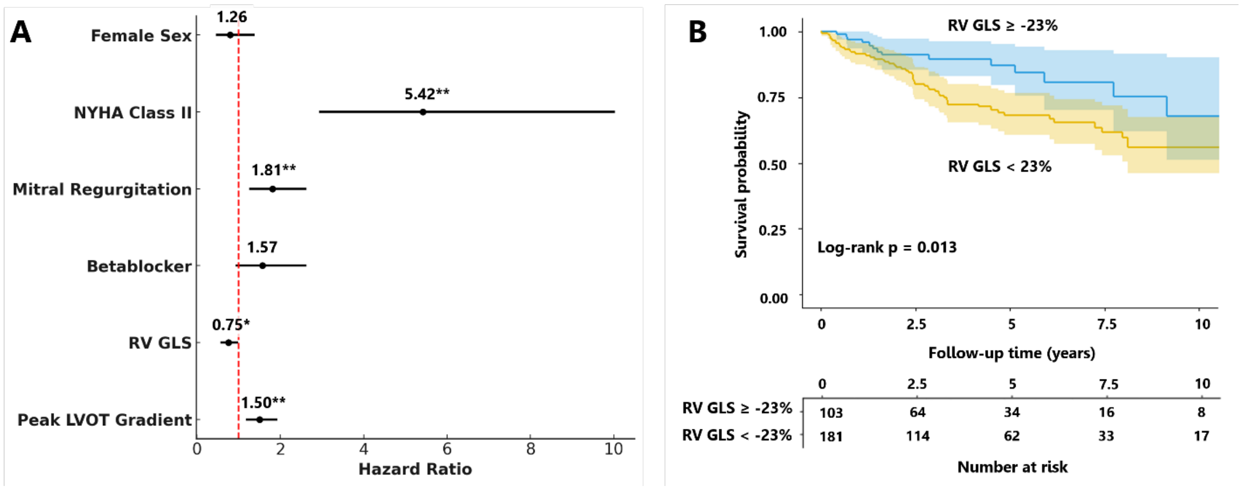

Figure 2. Predictors of heart failure events (A) and Survival plot stratified by right ventricular global longitudinal strain.

Forrest plots of multivariable Cox regression analysis and Kaplan–Meier curves for HF event–free survival, stratified by RV GLS (cut-off of -23%)

* $p<0.05$, ** $p<0.01$